2024-Winter Paper 3 – Subspecialities of Urology

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Q1. Discuss the diagnosis, staging and management of renal upper tract Urothelial tumor.

Type: LAQ (25 marks) (also answered in 2023-Winter Paper 2, Q1, with emphasis on aetiology)

Introduction

- UTUC (upper tract urothelial carcinoma) is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Peak age 70–90.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Concurrent bladder cancer in 17%.

Clinical features

- **Haematuria** (56–98%), **flank pain** (20–32%, dull from hydronephrosis or colic from clots); lumbar mass about 10%; incidental about 15%.
- Weight loss, bone pain or a mass suggest advanced disease.

Diagnosis

1. Cystoscopy: always, to exclude coexisting bladder cancer. Positive cytology with a normal bladder and urethra means the upper tracts must be examined.

2. Imaging

Test	Role and key points
CTU (computed tomography urography)	First line; highest accuracy (sensitivity ~92%, specificity ~95%). Radiolucent **filling defect**, hydronephrosis, non-visualisation. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU. Misses flat lesions (CIS, carcinoma in situ)
MRU (magnetic resonance urography)	If contrast or radiation is contraindicated; sensitivity ~75% for tumours <2 cm
Retrograde pyelography	Poorly filled calyces; accuracy ~75%; done with ureteroscopy
FDG-PET/CT (fluorodeoxyglucose positron emission tomography)	Nodes and metastases when contrast is contraindicated
Renogram	Split function; contralateral kidney must be assessed

- **Hydronephrosis** predicts invasive disease (up to 80% in ureteric tumours).

3. Urine cytology

- Sensitivity ~65%, grade-dependent (grade 1 ~20%, grade 3 ~75%). **Selective** ureteric washings are more accurate than voided urine. Take it before contrast. FISH (fluorescence in situ hybridisation) is optional for surveillance.

4. Flexible URS (ureteroscopy) with biopsy: the key test

- Adds accuracy (75–85% → ~90%). Biopsy gives **grade in ~90%**, but **stage cannot be assessed** (shallow samples; a Ta biopsy is often T3 at surgery).
- Indications: diagnostic doubt, planned kidney-sparing treatment, solitary kidney.
- Diagnostic URS before RNU (radical nephroureterectomy) raises bladder recurrence, without changing survival.

Staging (TNM 8th edition, 2017)

T	Definition
Ta / Tis	Non-invasive papillary / carcinoma in situ
T1	Subepithelial connective tissue
T2	Muscularis
T3	Pelvis: **peripelvic fat or renal parenchyma**; ureter: periureteric fat
T4	Adjacent organs, or **through the kidney into perinephric fat**

- **N1** single node ≤2 cm; **N2** node >2 cm or multiple. **M1** distant metastasis.
- **AJCC (American Joint Committee on Cancer) grouping:** stage I = T1, II = T2, III = T3; **any N+, T4 or M1 = stage IV**.
- **Grade:** WHO (World Health Organization) 2004/2016: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade.
- Regional nodes: pelvis and upper ureter → hilar and retroperitoneal; lower ureter → pelvic.

Risk stratification (EAU, European Association of Urology)

Low risk: all needed	High risk: any one
Unifocal, <2 cm	Multifocal or >2 cm
Low-grade cytology and biopsy	High-grade cytology or biopsy
No invasion on CTU	Hydronephrosis; invasion on imaging
—	Previous cystectomy; variant histology

- AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023 splits each group into favourable and unfavourable.
- **Prognostic factors:** stage (strongest), grade, LVI (lymphovascular invasion), nodes, sessile architecture, necrosis >10%, ureteric site, surgical delay >12 weeks.

Management

A. Kidney-sparing (low risk; imperative in solitary kidney, bilateral disease, CKD [chronic kidney disease])

- **Ureteroscopic laser ablation** (holmium or thulium) after biopsy; percutaneous resection for large pelvic tumours (tract seeding risk).
- **Segmental (distal) ureterectomy** with reimplantation or psoas hitch.
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); UGN-101 mitomycin gel for low grade. Lifelong endoscopic surveillance.

B. RNU with bladder cuff: standard for high risk

- Kidney outside Gerota's fascia; avoid opening the tract; remove the **intramural ureter and cuff**. Open, laparoscopic and robotic routes are equivalent.
- **Template lymphadenectomy** for high risk (hilar plus paracaval or para-aortic).
- **Single postoperative intravesical chemotherapy:** ODMIT-C (MMC 40 mg) cut bladder recurrence 28% → 16% at 1 year.

C. Perioperative chemotherapy

- **POUT (2020):** adjuvant platinum for pT2–T4 or N+ improved disease-free survival.
- Neoadjuvant platinum is preferred when post-RNU renal function would exclude cisplatin.

D. Metastatic: enfortumab vedotin + pembrolizumab (EV-302) first line; gemcitabine–platinum with avelumab maintenance; erdafitinib for FGFR3 alterations.

Follow-up

- After RNU: cystoscopy and cytology from 3 months; CTU yearly. After kidney-sparing: URS and CTU at closer intervals.

Diagram to draw: TNM levels of a renal pelvic tumour (T3 parenchyma versus T4 perinephric fat), beside the diagnostic flowchart CTU → cystoscopy → URS biopsy → risk group.

> **Key points to remember:**
>
> - CTU is the best test; URS biopsy gives grade, not stage.
> - Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
> - EAU low risk needs all favourable features; one adverse feature means high risk.
> - RNU + cuff + single MMC (ODMIT-C); adjuvant platinum (POUT).

Q2. Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

Type: LAQ (25 marks) (also answered in 2018-Summer Paper 1, Q1, with lymphatic anatomy)

Definition

- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in the urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (pyelolymphatic fistula).

Relevant lymphatic anatomy

- Intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction.
- Lumbar (para-aortic) trunks from the kidneys join the cisterna. Lymph flows cephalad and right to left.
- Valves normally stop backflow from the cisterna into renal hilar lymphatics.

Aetiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; Culex mosquito in urban India)	Congenital: lymphangiectasia, renal lymphangioma
Rarely other parasites	Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection])
	Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy

- Genital and urinary filariasis is almost always *W. bancrofti*; Brugia species spare the genitalia.

Pathogenesis

- **Life cycle:** a mosquito deposits infective larvae, which mature in central lymphatics over 6–9 months. Adults live 5–7 years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive theory:** adult worms and the host reaction cause **lymphangitis and fibrosis** of lymphatics between the intestinal trunks and thoracic duct.
- Lymphatics below the block **dilate**, their **valves fail**, and chyle **refluxes** into renal hilar and peripelvic lymphatics.
- Rising pressure ruptures the thin-walled lymphatics into a calyceal fornix, creating a **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Chyluria appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously.

Clinical features

- **Milky urine**, worse after fatty meals and on standing; may clear after fasting.
- **Chylous clots:** colic, retention.
- **Haematochyluria** (blood-stained chylous urine).
- **Systemic effects** of fat and protein loss: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- Endemic residence, filarial fever, hydrocele or lymphoedema.

- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Evaluation

1. Confirm chyle

- **Ether (chloroform) test:** milky urine shaken with ether clears, because the fat dissolves.
- **Urine triglycerides** positive; Sudan III staining shows fat globules; urine protein raised.
- Differentiate from **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial tests

- Night (midnight) blood smear: often negative by this stage.
- **Circulating filarial antigen** (card test), eosinophilia, PCR (polymerase chain reaction); ultrasound "filarial dance sign" in scrotal lymphatics.

3. Lateralise and localise the fistula

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice; bilateral in some.
- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy:** non-invasive; tracer seen in the kidney and bladder. MR lymphangiography or CT; pedal lymphangiography is the historical gold standard.

4. Baseline: albumin, haemoglobin, renal function.

Management

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Mild             -> diet + antifilarial drugs
Moderate/persist -> add endoscopic sclerotherapy (repeatable)
Severe or failed -> renal hilar lymphatic disconnection
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1. Conservative

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed directly into portal blood, not lacteals, so less chyle forms.
- **DEC (diethylcarbamazine)** 2 mg/kg three times daily, traditionally for 12 days (a 1-day course is as effective). It kills microfilariae but only modestly kills adults.
- **Doxycycline** (anti-*Wolbachia*, 200 mg daily for 4–8 weeks); albendazole; ivermectin.

2. Endoscopic sclerotherapy (renal pelvic instillation)

- Ureteric catheter on the affected side, head-down position. Instil **silver nitrate (0.5–1%)** or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- Good initial response; recurrence common.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:**
  1. Full renal mobilisation (nephrolysis).
  2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic.
  3. Strip the upper ureter; remove perinephric fat.
  4. Nephropexy; optional **omental wrap** around the hilum.
  5. On-table cystoscopy confirms clear efflux.

- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete.

4. Prevention

- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
> - Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
> - Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
> - Chyluria precedes elephantiasis and is often intermittent.

Q3a. Vascular Surgery in Urology.

Type: SAQ (10 marks) (vascular grafts answered in 2021-Summer Paper 3, Q2)

Assumed meaning: vascular procedures performed or met by the urologist (control and repair of vessels, caval surgery, transplant and renovascular surgery, penile vascular surgery and access), not vascular surgery as a specialty.

Principles

- **Control first:** proximal and distal control with vessel loops, Rummel tourniquets and atraumatic clamps (bulldog, Satinsky, DeBakey). Artery before vein.
- Systemic heparin before clamping; flush air and debris before tying; tension-free, everting anastomosis.
- **Sutures:** polypropylene (Prolene), double-armed. IVC (inferior vena cava) or aorta 3-0/4-0; renal vessels 4-0/5-0; transplant anastomoses 5-0/6-0.
- **Grafts:** autologous vein or artery for small or contaminated fields; PTFE (polytetrafluoroethylene) or Dacron for large vessels.

Applications

Setting	Vascular procedure
RCC (renal cell carcinoma) with IVC thrombus	Mayo levels 0–IV: level I milking and partial caval clamp; II full caval control; III liver mobilisation ± venovenous bypass; IV cardiopulmonary bypass ± deep hypothermic arrest. Patch cavoplasty if lumen <50%; caval replacement if wall invaded
Renal transplantation	Renal vein and artery end-to-side to external iliac vessels; Carrel patch; multiple arteries reconstructed on the bench
Renovascular disease	Ex vivo bench repair and **autotransplantation** (branch aneurysm, stenosis, ureteric loss). Angioplasty or stent first for fibromuscular dysplasia
Trauma	Main renal artery: endovascular stent; open repair only for solitary kidney or bilateral injury
PC-RPLND (post-chemotherapy RPLND) and sarcoma	IVC may be ligated or patched; aorta must be repaired or grafted
Penis	**Arterial revascularisation** (inferior epigastric artery to dorsal artery) for young men with post-traumatic pudendal stenosis; venous ligation investigational; priapism shunts (Grayhack saphenous–cavernosal)
Dialysis access	Arteriovenous fistula; PTFE graft if veins are poor

- **Endovascular:** angioembolisation for trauma, AML (angiomyolipoma) bleeding, pseudoaneurysm and AVF (arteriovenous fistula) after partial nephrectomy.

Complications

- Bleeding, thrombosis, stenosis, graft infection, pseudoaneurysm.

Diagram to draw: Mayo IVC thrombus levels with the matching vascular control.

> **Key points to remember:**
> - Proximal and distal control, heparin, flush, everting Prolene anastomosis.
> - Mayo levels decide caval control and bypass.
> - Bench surgery plus autotransplantation for complex renovascular lesions.

Q3b. Discuss the management of Metastatic Renal Cell Carcinoma.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 2, Q1; immunotherapy in detail in 2024-Winter Paper 2, Q3e)

Assessment

- CT chest, abdomen and pelvis; bone scan or brain MRI if symptomatic. **Biopsy** for histology.
- **IMDC (International Metastatic RCC Database Consortium) risk factors:** Karnofsky <80%; haemoglobin < normal; calcium > ULN (upper limit of normal); neutrophils > ULN; platelets > ULN; <1 year from diagnosis to systemic therapy.
- Favourable 0, intermediate 1–2, poor ≥3; median survival 43.2, 22.5 and 7.8 months.

First-line systemic therapy (clear cell RCC; NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

IMDC risk	Preferred	If IO (immuno-oncology) not possible
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib, pazopanib; active surveillance for slow, low-volume disease
Intermediate/poor	Same three, **or nivolumab + ipilimumab**	Cabozantinib, sunitinib, pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss → HIF (hypoxia-inducible factor) → VEGF (vascular endothelial growth factor), the target of TKIs (tyrosine kinase inhibitors). Checkpoint inhibitors release T-cell brakes.
- **Immunotherapy** trials (CheckMate 214, KEYNOTE-426, CheckMate 9ER, CLEAR) and toxicity: see 2024-Winter Paper 2, Q3e.

Later lines: switch mechanism

- After IO: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor; LITESPARK-005).
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR).

Cytoreductive nephrectomy (CN)

- **CARMENA (2018):** sunitinib alone non-inferior to CN + sunitinib in intermediate/poor risk. **SURTIME:** deferred CN safe.
- Now: immediate CN only for fit, low-burden patients; **deferred CN** in responders; avoid in poor risk. Palliative nephrectomy or embolisation for bleeding.

Metastasectomy and radiotherapy

- Complete resection of solitary or few metastases: 5-year survival 35–50%.
- **SBRT (stereotactic body radiotherapy)** for oligometastases; radiosurgery for brain; palliative radiotherapy for bone pain.

Supportive and special groups

- Zoledronic acid or denosumab for bone metastases. Non-clear cell: trials, cabozantinib or sunitinib. Adjuvant pembrolizumab after complete metastasectomy (KEYNOTE-564).

Diagram to draw: flowchart: biopsy + IMDC → IO combination → deferred CN in responders → switch class on progression.

Key points to remember:

- IMDC 6 factors; poor = ≥3.
- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor.
- CARMENA and SURTIME: no routine upfront CN.
- Belzutifan after IO and TKI.

Q3c. Prognostic indicators in emphysematous pyelonephritis.

Type: SAQ (10 marks)

Definition

- EPN (emphysematous pyelonephritis) is an acute **necrotising** infection of the renal parenchyma and perirenal tissue by **gas-forming organisms**. It is a urological emergency.
- Most patients are **diabetic**; obstruction in about a quarter.
- Organisms: *E. coli* (commonest), *Klebsiella*, *Proteus*. Proposed mechanism: high tissue glucose fermented to CO_2 in poorly perfused tissue.

Imaging-based prognostic classifications

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only (emphysematous pyelitis)
2	Renal parenchyma, no extrarenal extension
3A	Extension into **perinephric** space
3B	Extension into **pararenal** space
4	**Bilateral** EPN or EPN in a **solitary kidney**

2. Wan classification (CT pattern)

Type	CT	Mortality
I	Streaky or mottled gas, **no fluid**; rapid parenchymal destruction	50–60%
II	Renal or perirenal **fluid**, bubbly or loculated gas, or gas in collecting system	<20%

- **Why:** fluid means a walled-off collection that can be drained. Streaky gas without fluid means necrosis faster than the host can contain it.

Clinical and laboratory prognostic factors

- **Huang–Tseng risk factors: thrombocytopenia**, **AKI (acute kidney injury)/raised creatinine**, **altered sensorium**, **shock**.
- **Other adverse factors:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

How prognosis guides management

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All: resuscitation, glucose control, IV antibiotics, relieve obstruction (stent/PCN)
Class 1–2               -> antibiotics + PCD
Class 3–4, <2 risk factors -> antibiotics + PCD (usually succeeds)
Class 3–4, >=2 risk factors or PCD failure -> nephrectomy
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- PCD (percutaneous catheter drainage); PCN (percutaneous nephrostomy).
- Reported mortality 19–43%; routine emergency nephrectomy still carried 40–50%, so drainage-first care is now standard.
- **Also know:** emphysematous pyelitis (gas in the collecting system only) is often non-diabetic and responds to antibiotics.

Diagram to draw: CT sketches of Huang–Tseng classes 1, 2, 3A/3B and 4.

Key points to remember:

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors.

Q3d. Mirabegron in over active bladder.

Type: SAQ (10 marks)

Introduction

- **Mirabegron** is the first **β3-adrenoceptor agonist** for OAB (overactive bladder: urgency, with or without urgency incontinence, usually with frequency and nocturia). FDA (Food and Drug Administration) approval 2012.

Mechanism of action

- **β3 receptors** are the most highly expressed adrenoceptor in human detrusor.
- Activation → adenylyl cyclase → **↑ cAMP (cyclic adenosine monophosphate)** → detrusor relaxation during **storage**.
- Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR (post-void residual) barely rises.

Pharmacology and dose

- Oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks.
- **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.

Efficacy

- Phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Combination with solifenacin** (SYNERGY, SYNERGY II, BESIDE): better than either alone, without extra cardiovascular effects.

Adverse effects and contraindications

- Hypertension, tachycardia, headache, nasopharyngitis, UTI (urinary tract infection).
- **Dry mouth and constipation near placebo rates**, far below antimuscarinics.
- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia and BOO (bladder outlet obstruction) with antimuscarinics (retention).

Place in therapy

- After behavioural therapy, as first-line pharmacotherapy alongside antimuscarinics (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]).
- **Preferred in the elderly and cognitive impairment:** no anticholinergic burden. Antimuscarinic use >3 months is linked to 10–65% more new-onset dementia (SUFU white paper).
- Add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

Diagram to draw: detrusor cell: noradrenaline → β3 → cAMP → relaxation, versus acetylcholine → M3 → contraction.

Key points to remember:

- β3 → cAMP → storage-phase relaxation; voiding spared.
- 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Preferred over antimuscarinics in the elderly (dementia risk).
- Add-on to solifenacin (SYNERGY) works.

Q3e. Management of stress urinary incontinence.

Type: SAQ (10 marks)

Definition and mechanism

- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing (ICS, International Continence Society).
- Two mechanisms: **urethral hypermobility** (loss of the suburethral "hammock") and **ISD (intrinsic sphincter deficiency)**.

Evaluation (brief)

- Bladder diary, pad test, **cough stress test**, PVR (post-void residual), urinalysis; Q-tip >30° = hypermobility.
- **Urodynamics not needed** in the uncomplicated (index) patient (VALUE trial); use if mixed, recurrent, neurogenic or with prolapse.

Female SUI

1. Conservative (first line)

- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT (pelvic floor muscle training)** for at least 3 months (Cochrane: more cure than no treatment).
- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not USA.

2. Surgery

Operation	Principle	Key facts
MUS (mid-urethral sling): retropubic (TVT, tension-free vaginal tape) or transobturator (TOT)	Tension-free tape acts as a new pubourethral ligament	First line for index SUI. **TOMUS:** 80.8% vs 77.7% cure (equivalent). Retropubic: more bladder perforation; transobturator: more groin pain
Single-incision sling	Short tape, vaginal only	Option; non-inferior to TOT in most studies
Burch colposuspension (open or laparoscopic)	Paravaginal fascia to Cooper's ligament; restores bladder neck position	85–90% at 1 year; for women declining mesh
Autologous pubovaginal sling (rectus fascia)	Bladder-neck support and compression	ISD, recurrent SUI, after mesh removal. **SISTEr:** sling 47% vs Burch 38%, more voiding dysfunction
Urethral bulking agents	Submucosal coaptation	Frail patients; less durable
AUS (artificial urinary sphincter)	Circumferential cuff	Severe refractory ISD

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (~2%), pain.

Male SUI (brief)

- Usually after radical prostatectomy. PFMT first; no surgery for 6–12 months, as continence keeps improving.
- **AUS** for moderate–severe or irradiated men (gold standard, ~80% continent); **male sling** for mild–moderate leakage without radiotherapy. Bulking agents not advised.

Diagram to draw: sagittal pelvis showing retropubic and transobturator tape paths and Burch sutures.

Key points to remember:

- PFMT first; MUS first-line surgery.
- TOMUS: retropubic = transobturator; SISTEr: fascial sling > Burch.
- VALUE: no urodynamics in index SUI.
- Men: AUS gold standard; sling for mild, non-irradiated.

Q3f. Sting procedure.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3c)

Taken as STING (subureteric Teflon injection), the endoscopic treatment of VUR (vesicoureteral reflux).

Definition and principle

- Endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports the intramural ureter and restores the **flap-valve**.
- Popularised by **O'Donnell and Puri, 1986**.

Bulking agents

Agent	Particle	Problem
Teflon (PTFE, polytetrafluoroethylene paste)	90% <40 µm, permanent	**Migration**; never FDA (Food and Drug Administration) approved
Dx/HA (dextranomer/hyaluronic acid, Deflux)	**80–250 µm**; gel resorbed	Volume loss about 23%; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Late ureteric obstruction

Indications

- Persistent primary VUR (grades II–IV) with breakthrough febrile UTI (urinary tract infection), new scars or parental preference; residual VUR after reimplantation.
- **Relative contraindications:** grade V, **UVJ (ureterovesical junction) obstruction**, untreated BBD (bladder and bowel dysfunction).

Technique

1. Antibiotic at induction; offset-lens cystoscope; 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (implant extrudes).

Technique	Needle entry	Result
STING	Below orifice	Baseline
HIT (hydrodistension implantation technique)	Inside hydrodistended orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Results

- Low-grade VUR about 90% after one injection. Elder 2006 meta-analysis: grade I–II about 79%, III 72%, IV 63%, V 51%.
- **Swedish Reflux Trial:** 20% recurrence at 2 years.

Complications

- Failure (repeat 1–2 injections, then reimplant); contralateral VUR; **early or late obstruction**; calcified mound mimics a stone.

Follow-up

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound ± VCUG (voiding cystourethrogram) at 3 months.

Diagram to draw: needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

> **Key points to remember:**
>
> - O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
> - Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
> - Watch late obstruction; calcified mound mimics a stone.

Rapid Revision Sheet – 2024-Winter Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss the diagnosis, staging and management of renal upper tract Urothelial tumor. (LAQ)

- CTU is the best test; URS biopsy gives grade, not stage.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features; one adverse feature means high risk.
- RNU + cuff + single MMC (ODMIT-C); adjuvant platinum (POUT).

Q2. Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

Q3a. Vascular Surgery in Urology. (SAQ)

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.
- Mayo levels decide caval control and bypass.
- Bench surgery plus autotransplantation for complex renovascular lesions.

Q3b. Discuss the management of Metastatic Renal Cell Carcinoma. (SAQ)

- IMDC 6 factors; poor = ≥3.
- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor.
- CARMENA and SURTIME: no routine upfront CN.
- Belzutifan after IO and TKI.

Q3c. Prognostic indicators in emphysematous pyelonephritis. (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors.

Q3d. Mirabegron in over active bladder. (SAQ)

- β3 → cAMP → storage-phase relaxation; voiding spared.
- 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Preferred over antimuscarinics in the elderly (dementia risk).
- Add-on to solifenacin (SYNERGY) works.

Q3e. Management of stress urinary incontinence. (SAQ)

- PFMT first; MUS first-line surgery.
- TOMUS: retropubic = transobturator; SISTEr: fascial sling > Burch.
- VALUE: no urodynamics in index SUI.
- Men: AUS gold standard; sling for mild, non-irradiated.

Q3f. Sting procedure. (SAQ)

- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.

- Watch late obstruction; calcified mound mimics a stone.